

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2013

Time: 2 hour

250381

MATHEMATICS

READ THESE INSTRUCTIONS CAREFULLY

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 questions
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not on this question booklet.

HOW TO USE THE ANSWER SHEET

4. Use only ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number (i.e. School Number and three figure candidate's Number) in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A, B, C and D. In each case only **ONE** of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen in written.

Example

In the question booklet:

11. What is the value of $\frac{6(24-18)}{6} + 6 \times 4$?

6

- A. 30
- B. 25
- C. 10
- D. 28

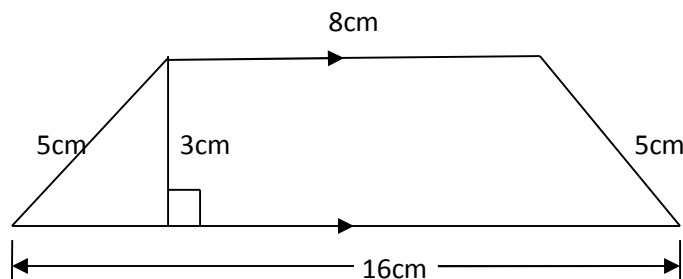
The correct answer is C(10)

On the answer sheet:

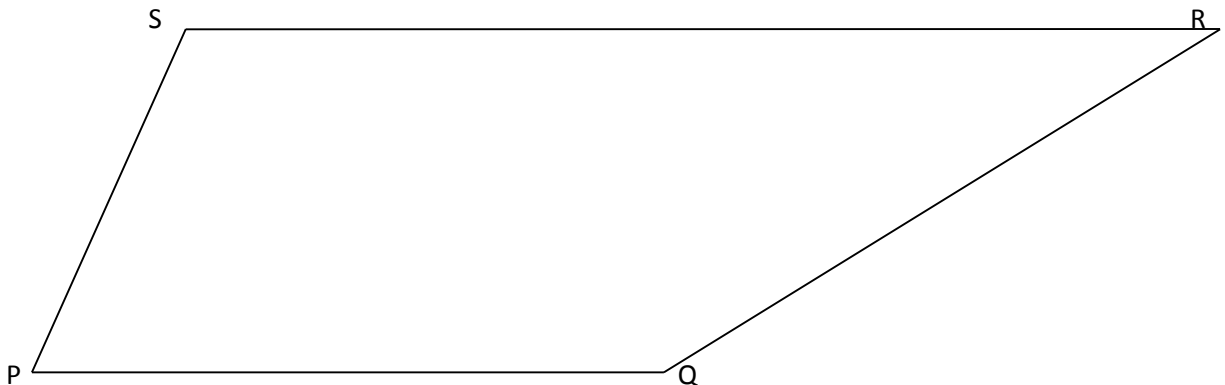
1 (A) (B) (C) (D) 11 (A) (B) (C) (D) 21 (A) (B) (C) (D) 31 (A) (B) (C) (D)

12. Your dark line **MUST** be within the box.
 13. For each question **ONLY ONE** box is to be marked in each set of four boxes.
- The Kenya National Examinations Council, 2013

1. Which one of the following number is one hundred and two thousand and eight hundredths?
 - A. 100200.08
 - B. 102000.08
 - C. 102.08
 - D. 102000.80
2. What is value of $45 + 3(8 - 2^2) - 7 \times 6 \div 3$?
 - A. 178
 - B. 51
 - C. 49
 - D. 43
3. What is the total value of digit 7 in the number 42765831?
 - A. 7 000
 - B. 70 000
 - C. 700 000
 - D. 7 000 000
4. What is the value of $3(0.6^2 - 0.2^2) + 2.4 \div 0.8$?
 - A. 3.96
 - B. 4.04
 - C. 4.2
 - D. 5.4
5. Which one of the ratios given below represents 60%?
 - A. 3:8
 - B. 3:5
 - C. 5:3
 - D. 5:8
6. How many days were there from 1st January to 31st March, 2012?
 - A. 89
 - B. 90
 - C. 91
 - D. 92
7. What is the area of the figure shown below?



- A. 24cm^2
 B. 36cm^2
 C. 34cm^2
 D. 40cm^2
8. What is 1844.056 when rounded off to two decimal places?
 A. 1844.05
 B. 1844.00
 C. 1844.10
 D. 1844.06
9. The fraction $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$ and $\frac{4}{9}$ are to be arranged in order from the smallest to the largest. Which one of the following is the correct order?
 A. $\frac{2}{3}$, $\frac{3}{4}$, $\frac{4}{9}$, $\frac{5}{6}$
 B. $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$, $\frac{4}{9}$
 C. $\frac{4}{9}$, $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$
 D. $\frac{5}{6}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{4}{9}$
10. What is the difference between the L.C.M and the H.C.F. of the number 18, 24 and 72?
 A. 6
 B. 66
 C. 72
 D. 78
11. What is the value of
 $(4\frac{1}{2} - 2\frac{1}{4} + 1\frac{1}{2} \div \frac{3}{2}) \div 2$?
 A. $\frac{21}{4}$
 B. $\frac{11}{4}$
 C. $\frac{3}{14}$
 D. $\frac{15}{8}$
12. In the quadrilateral PQRS below, construct a perpendicular from S to meet line PQ at T. Draw a line parallel to TS passing through Q to meet line SR at V. Join TV.



What is the length of line TV in cm?

- A. 2.3
- B. 4.0
- C. 5.0
- D. 6.4

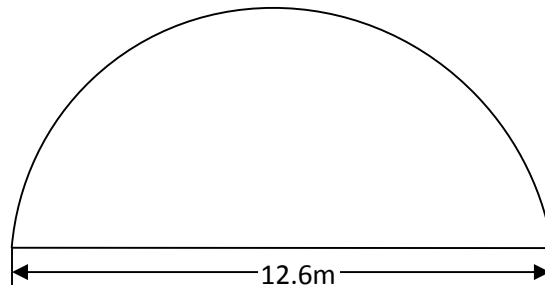
13. Wandere used $\frac{1}{8}$ of his plot of land as homestead, $\frac{1}{3}$ for flowers, $\frac{1}{4}$ for nappier grass and the remainder for vegetables. If the area for vegetables was 280cm^2 what was the area for flowers?

- A. 120m^2
- B. 240m^2
- C. 32m^2
- D. 960m^2

14. Beka harvested 6 bags of beans and 8 bags of maize. He sold each bag of beans at sh 4 800 and each bag of maize at 1 500. How much money was left after paying sh 24 500 for school fees?

- A. Sh 4 300
- B. Sh 16 300
- C. Sh 48 800
- D. Sh 65 300

15. The figure below represents a flower garden



The garden is to be fenced all around using strands of wire. What length of the wire, in metres, is required? (Take $\pi \frac{22}{7}$)

- A. 208.8
- B. 129.6
- C. 79.2
- D. 32.4

16. On the certain day a shopkeeper sold 6 rulers, 15 rubbers, 24 pencils, and 9 exercise books. If these items were to be represented in a pie chart, what angles would represent the number of exercise books?

- A. 160°
- B. 100°
- C. 60°

D. 40^0

17. What is the value of x in the equation

$$\frac{2}{3}(2x + 4) + 6 = 10?$$

A. 5

B. 4

C. 2

D. 1

18. Chebet left town P at 7:00am and travelled to town R, a distance of 320km via town Q. She took two hours to travel from town P to town Q at an average speed of 85km/h. After a 45 minutes rest she continued with the journey at an average speed of 75km/h. At what time did she arrive at town R?

A. 11.45am

B. 11.00am

C. 9.45am

D. 9.00am

19. A lorry which carried four people of average mass 80.5kg and also loaded with cabbages weighed 5.8 tonnes . What was the mass in tonnes when empty?

A. 2 400

B. 2 319.5

C. 2 078

D. 322

20. The table below shows the number of candidates who sat for a certain examination in the years 2008 and 2009.

GENDER	YEAR	
	2008	2009
Girls	328652	345 454
Boys	367125	381600

How many more candidates sat for the examination in 2009 than 2008?

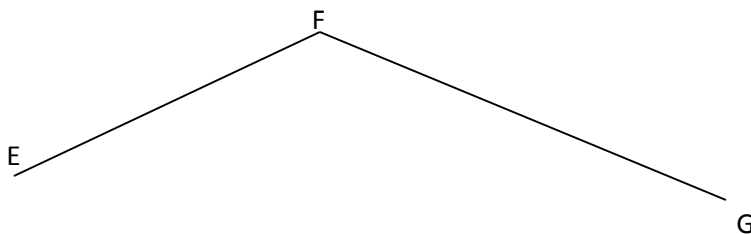
A. 74619

B. 31277

C. 16802

D. 14475

21. What is the size of the obtuse angle EFG drawn below?



- A. 50^0
- B. 130^0
- C. 133^0
- D. 230^0

22. What is the value of

$$\frac{\sqrt{5+1}}{(1-3)}$$

- A. $5/64$
- B. $5/8$
- C. $1\frac{3}{4}$
- D. $3\frac{1}{8}$

23. Figure PQRS shown below is parallelogram. Line RST is a straight line.

What is the size of angle TQR?

- A. 140^0
- B. 110^0
- C. 40^0
- D. 30^0

24. A salesman is paid monthly salary of sh 6 000 and a 5% commission on goods he sales above sh 20 000 each month. In a certain month the salesman eared sh 13 000. How much money was the sales of goods that month?

- A. Sh 120 000
- B. Sh 140 000
- C. Sh 160 000
- D. Sh 280 000

25. The diagram below represents a cylindrical solid of diameter 56cm. The solid has circular hole of radius 7cm. The length of the solid is 120cm.

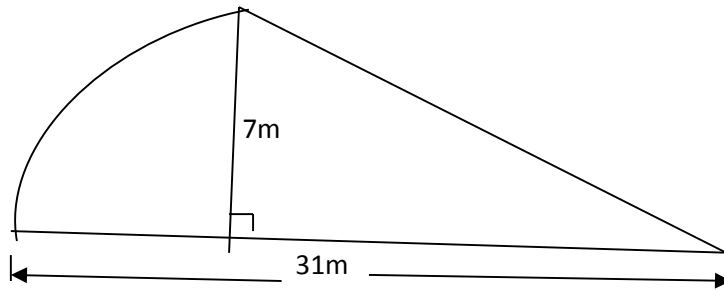
What is the volume of the solid?(Take $\pi = 22/7$)

- A. 18480cm^3
- B. 295680cm^3
- C. 314160cm^3
- D. 277200cm^3

26. In a game park tourist saw giraffes, elephants and lions. The number of lions seen was n. The number of elephant seen was three times the number of lions. The number of elephant seen was six less than the number of giraffes. How many animals altogether did the tourist see in the park?

- A. $7n + 6$
- B. $3n + 12$
- C. $7n - 6$
- D. $n + 9$

27. The figure below represents a flower garden in the shape of a quarter of the circle of radius 7m and a right angled triangle.



What is the area of the flower garden? (Take $\pi = 22/7$)

- A. 95m^2
 - B. $108 \frac{1}{2} \text{m}^2$
 - C. $122 \frac{1}{2} \text{m}^2$
 - D. 147m^2
28. Khalifa bought 5 trays of eggs in one of the trays of eggs @ sh 200. All eggs in one trays broke and he sold the remaining trays @ sh 210. What percentage loss did he make?
- A. 5%
 - B. 84%
 - C. $19 \frac{1}{21}\%$
 - D. 16 %
29. Six men working at the same rate, can complete digging a shamba in 8 days. How many days will it take 4 of the men to complete the same work?
- A. 12
 - B. $5\frac{1}{3}$
 - C. 4
 - D. 3
30. Mwangeli receives a salary of sh 15 000 per month. In a certain month she spent 0.2 of the salary on the rent, 0.3 on food and 0.1 on transport. She spent 0.25 Of the remaining on school fees and saved the rest. How much money did she saved in the month ?
- A. Sh 4 500
 - B. Sh 3 750

- C. Sh 2 250
D. Sh 1 500

31. The figure below shows a scale drawing for a route of a map for a bicycle race. The scale used is 1 :200 000.

Atieno was in the bicycle race, Her bicycle down at check point 2 and had to retire from the race. How many kilometers was she away from the finish point when she retired

- A. 40km
B. 16km
C. 24km
D. 10km

32. Sangale left home at 2330h on Monday for a journey which took $21\frac{1}{4}$ days to complete. On what day and time in am/pm system complete the journey?

- A. Thursday 5.30am
B. Wednesday 5.30am
C. Thursday 11.30pm
D. Wednesday 11.30pm

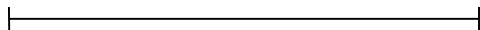
33. The table below shows price of food for a hotel.

Table showing price of food in shillings

	Ugali	Mchele	Chips
Nyama	150	180	200
Samaki	160	170	210
Kuku	240	180	250

- A. Sh 480
B. Sh 600
C. Sh 2 270
D. Sh 2 250

34. On line PQ drawn below construct a triangle NPQ such that angle NPQ=30° and line PN =10cm bisect angle PQR and draw the bisector to meet PN at R.



What is the size of angle NRS?

- A. 95°
B. 85°

C. 57^0 D. 43^0

35. The cash price of a cupboard is sh 3 000. An interest of 25% is charged if the cupboard is bought on hire purchase terms. Amina bought the cupboard on hire purchase terms. She paid a deposit and twelve equal instalments. Each instalment was sh 262.50

How much deposit did she pay?

A. Sh 3 750

B. Sh 750

C. Sh 600

D. Sh 150

36. The table below shows a timetable for a certain airline plying Nairobi –Kisumu-Mombasa route.

From	To	Flight	Departure	Arrival
Nairobi	Mombasa	209	0826h	0900h
Mombasa	Nairobi	210	1430h	1500h
Nairobi	Kisumu	211	1645h	1730h
Kisumu	Nairobi	212	1845h	1930h

What is the difference between the time taken by flight No.212 to fly between Kisumu and Nairobi, and flight No. 210 to fly between Mombasa and Nairobi?

A. 15min

B. 30min

C. 45min

D. 75min

37. What is the value of $2a + b$ when $c = 4$, $b = 2c + 1$ and $a = b + 2$?

A. 3

B. $\frac{43}{4}$

C. 7

D. $\frac{73}{4}$

38. A cylindrical water tank has a diameter of 1.4m and a height of 4m. How many litres of water will it hold when full? (Take $\pi = \frac{22}{7}$)

A. 6.16

B. 61.6

C. 616

D. 6160

39. The mean of the following marks for pupils in a certain class is 72.

66,86,63,x,79,69,85,63,88

What is the medium mark?

- A. 74
- B. 70
- C. 63
- D. 41

40. A trader deposited sh 20 000 in a bank. The money was in the following denominations:

Sh 1000 – 5 notes

sh 500 – 15 notes

Sh 200 – 24 notes

The remaining amount was n

100 shillings and 50 shillings note. The 100 shillings and 50 shillings notes were equal in numbers. How much money did the trader deposit in 50 shillings notes?

- A. Sh 2 700
- B. Sh 1 800
- C. 1 300
- D. 900

41. A cylindrical container has a radius of 6.3cm and is 15cm high. Its curved surface is completely covered with a label. What is the area of the label in cm^2

(Take $\frac{22}{7}$)

- A. 297.00
- B. 594.00
- C. 843.48
- D. 1871.10

42. The graph below shows a journey by bus from town X to town Y and back.

What was the average speed of the bus the whole journey?

- A. 38.40km
- B. 19.20km
- C. 50.53km
- D. 40.00km

43. Wanja deposited sh 24 600 in a certain financial institution which paid a simple interest per annum. After 9 months the total amount of the money plus interest earned was sh 25 338. At the money earned interest?

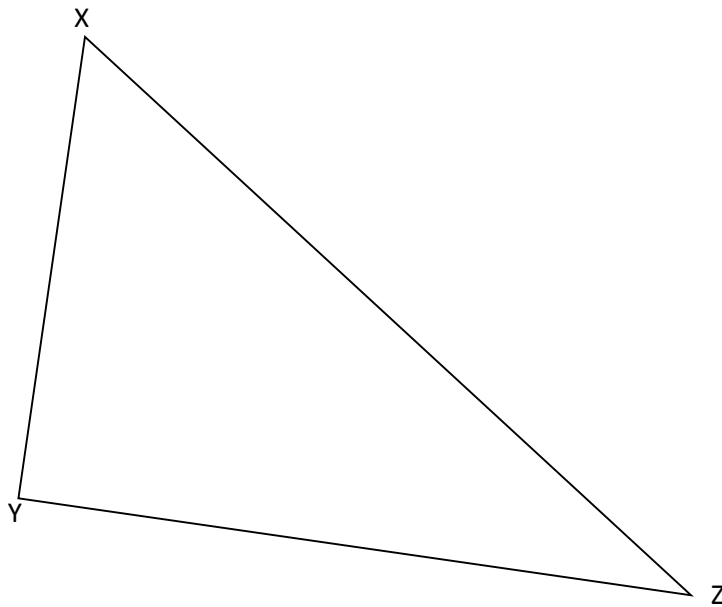
- A. 4%
- B. 3%
- C. $2\frac{1}{3}\%$
- D. $\frac{1}{3}\%$

44. In the figure shown below, line AB is parallel to line CD. Lines EC, EF and BF are transversals. Which one of the statements below is correct?

- A. $q + s = 1800$

- B. $p + r = t + v$
- C. $v + u = s + r$
- D. $q + t = 1800$

45. Construct a circle that touches the three sides of the triangle XYZ given below.



What is the diameter of the circle?

- A. 8.6cm
 - B. 4.3cm
 - C. 3.8cm
 - D. 1.9cm
46. Perimeter of a triangle is 32cm. Two sides of the triangle measures 10cm and 12cm. What is the area of the triangle?
- A. 24
 - B. 40
 - C. 48
 - D. 50
47. The price of an article in a shop was increased by 10%. The price was decreased by 20% The price of the article after the decrease was sh 79 200.
What was the original price before the increase ?
- A. Sh 72 00
 - B. Sh 88 000
 - C. 90 000
 - D. 99 000
48. The figure below shows a pattern of shapes.

Which one of the following shapes is the next in the pattern?

49. The following are length of sides of different triangles

- (i) 5, 12, 13
- (ii) 7, 24, 25
- (iii) 5, 11, 13
- (iv) 3, 4, 5

Which one of the choices below contain all right angled triangles?

- A. (i) (ii) (iii)
- B. (i) (iii) (iv)
- C. (ii) (iii) (iv)
- D. (i) (ii) (iv)

50. Which one of the following figures is the net of a triangular prism?